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Integration of Cancer Genomics Data for Tree-Based Dimensionality Reduction and Cancer Outcome Prediction

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Abstract: Accurate outcome prediction is crucial for precision medicine and personalized treatment of cancer. Researchers have found that multi-dimensional cancer omics studies outperform each data type (mRNA, microRNA, methylation or somatic copy number alteration) study in human disease research. Existing methods leveraging multiple level of molecular data often suffer from various limitations, *e.g.*, heterogeneity, poor robustness or loss of generality. To overcome these limitations, we presented the tree-based dimensionality reduction approach for the identification of smooth tree graph and developed accurate predictive model for clinical outcome prediction. We demonstrated that 1) Discriminative Dimensionality Reduction via learning a Tree (DDRTree) achieved reduced dimension space tree with statistical significance; 2) Tree based support vector machine (SVM) classifier improved prediction performance of cancer recurrence as compared to *t*-test based SVM classifier; 3) Tree based SVM classifier was robust with regard to the different number of multi-markers; 4) Combining multiple omics data improved prediction performance of cancer recurrence as compared to a single-omics data; and 5) Tree based SVM classifier achieved similar or better prediction performance when compared to the features from state-of-the-art feature selection methods. Our results demonstrated great potential of the tree-based dimensionality reduction approach based clinical outcome prediction.

Keywords: clinical outcome prediction · tree-based dimension reduction · Integration of cancer genomics data · support vector machine

1 Introduction

Cancer prognosis is one of the leading areas of study in clinical cancer research. Accurate outcome prediction typically helps to refine the prognostic stratification by risk of cancer recurrence, determine the appropriate therapy and thus improve survival rate of cancer patients. In particular, large-scale cancer genomics project, such as The Cancer Genome Atlas (TCGA), yields an overwhelming amount of cancer genomics, transcriptomic, epigenomic and proteomic data, which have been applied to the discovery of somatic genomic alterations,[1] the recognition of biologically discrete molecular subsets and pathways,[2] the identification of clinically and phenotypically significant network modules with disease mutations,[3] the stratification of molecular subtypes,[4] and the development of support vector machine algorithm to predict the patient responses to chemotherapeutic drugs.[5] However, to our knowledge, methods that could demonstrate the potential clinical utility of diverse molecular data in aggregate are still lacking. Therefore, there is an urgent need to develop accurate computational approach that connects multiple types of molecular data for clinical outcome prediction.

Based on a single-omics data from different types of cancer, several studies have reported a gene-expression signature as a predictor of survival in breast cancer,[6] low density separation approach based gene-expression data that predicts clinical outcome in colorectal cancer patients,[7] copy number alteration burden that is prognostic for recurrence and metastasis in prostate cancer,[8] long noncoding RNAs (lncRNAs) expression correlated with recurrent mutations, clinical features, and clinical outcome in acute myeloid leukemia,[9] and circulating microRNA signature for the diagnosis of very high-risk prostate cancer.[10] Notably, multi-dimensional genomic studies have been applied to clinical outcome prediction and outperformed each data type (mRNA, microRNA, methylation or somatic copy number alteration) study. A graph-based integration framework is proposed for clinical outcome prediction using copy number alteration, methylation, miRNA and gene expression data.[11] On the other hand, integration of somatic mutation, expression and functional data identifies clinically relevant potential driver genes in breast cancer.[12] Integration of multi-omics data and genomic knowledge also develops understanding of molecular pathogenesis and underlying biology in cancer, and thus improves power in predicting clinical outcomes.[13] Moreover, a methodological framework integrates multi-omics data and biological knowledge to determine meta-dimensional knowledge-driven genomic interactions associated with clinical outcomes of cancer.[14]

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The computational difficulty is associated with the estimation in the high-dimensional and low-sample size settings in multiple types of omics measurements studies, raising questions about dimensionality reduction analysis. Dimensionality reduction methods have been widely used for high-dimensional data, which not only reduce the computational complexity but also improve the significance of learning models. A variety of linear projections techniques have been widely used for the representation of high dimensional data, such as Principal Component Analysis (PCA),[15] Linear Discriminant Analysis (LDA),[16] Multidimensional Scaling (MDS)[17] and Locality Preserving Projection (LPP).[18] Importantly, some manifold learning algorithms such as ISOMAP,[19] Locally Linear Embedding (LLE),[20] and Laplacian Eigenmaps[21] have been applied to a data set for conducting nonlinear dimensionality reduction and thus found a lower dimensional manifold. Graphs are powerful tools for identifying the intrinsic structure information hidden in data, and thus graph-based dimensionality reduction algorithms have been applied to achieve better performance for face recognition[22] and single-cell RNA-seq data.[23] Specifically, some graph cut criteria, e.g., ratio-cut,[24] normalized-cut (Ncut),[25] and min-max-cut,[26] proved to be robust and adaptable separability criteria. Given the complex geometry of tree-structured data spaces, many progress have been made in the development of distance metrics for parameterization of data analysis. Among them, the Tree Bisection and Reconnection (TBR) distance,[27] the Tree Edit (TED) distance,[28] and the Quotient Euclidean (QED) distance[29] have been applied to tree-structured data parameterization. More specifically, the T-A matrix based clustering (TAMBAC) framework is developed for tree-structured data clustering based on a novel tree data parameterization (called TAMP) and the structure-constrained nonnegative matrix factorization (SCNMF).[30] TAMP represents both the topological and geometric information of trees in forms of matrices so that the tree-structured data analysis problem could be solved in the matrix manifold realm. The performance demonstrates the efficacy and accuracy of the TAMBAC framework. The prognosis model with the clinical features has showed great potential in clinical outcome prediction for different types of cancers. The DREAM Challenge developed a promising prognosis model for improving the prediction of survival in metastatic castration-resistant prostate cancer (mCPRC)[31] and constructed routinely collected clinical features based predictive model for identifying patients with mCRPC who are likely to discontinue treatment because of adverse events.[32] Specifically, the partial least squares method was used for tumor classification, in which the response variables Y is represented either as a dummy matrix[33] or as a pseudo-response variable.[34]

In this paper, we presented tree-based dimensionality reduction approach for multiple molecular data across two types of cancer and developed SVM classifier for cancer recurrence prediction. We used genome-wide gene expression, microRNA expression and somatic copy number alteration profiles to classify the glioblastoma (GBM) and lung squamous cell carcinoma (LUSC) datasets into the "recurrence" and "non-recurrence" groups respectively. We demonstrated that tree based SVM classifier improved prediction performance of cancer recurrence as compared to *t*-test based SVM classifier. We discovered that tree based SVM classifier was robust with regard to the different number of multi-markers. We then investigated that combining multiple omics data yielded superior prediction performance of cancer recurrence as compared to a single-omics data. We also compared our approach with state-of-the-art feature selection methods and achieved comparable or superior performance.

2 Materials and Methods

2.1 Patient Cohort Characteristics

The gene expression and microRNA datasets in cancer patients were retrieved from the Pan-cancer project (syn300013) on Synapse respectively (http://www.synapse.org) and somatic copy number alteration (SCNA) profile was obtained from Firehose (https://confluence.broadinstitute.org/display/GDAC/Home). We downloaded the clinical variables from TCGA data portal (https://tcga-data.nci.nih.gov/tcga/) and overall survival data from Firehose. We collected 210 GBM patients and 121 LUSC patients with multi-omics datasets, clinical information and overall survival time describing the disease at surgery and the clinical course of the disease. For GBM, the platform was Agilent 244 K Custom Gene Expression G4502A for mRNA expression, Agilent 8×15 K Human miRNA-specific microarray for miRNA expression and Affymetrix Genome-Wide Human SNP Array 6.0 for SCNA respectively. For LUSC, the platform was Illumina HiSeq 2000 RNA Sequencing V2 for mRNA expression, Illumina Genome Analyzer/HiSeq 2000 miRNA sequence for miRNA expression and Affymetrix Genome-Wide Human SNP Array 6.0 for SCNA respectively. The mRNA expression, miRNA expression and SCNA were standardized using z-score transformation. For each cancer type, we assigned the cancer patients as the "recurrence" cohort if recurrence was observed within 3 years of follow-up and the "non-recurrence" cohort if recurrence was not observed within 3 years of follow-up. The multi-omics datasets of tumor samples with GBM and LUSC are listed in Table 1.

2.2 Formulation of DDRTree

Discriminative Dimensionality Reduction via learning a Tree (DDRTree)[35] was used for dimensionality reduction and formulated as following three parts.

Table 1. Summary of datasets used for model development and validation. R and NR stand for Recurrence and Non-Recurrence in the data respectively. SCNA stands for Somatic Copy Number Alteration.

Tissue	Number of samples (R/NR)	Training dataset (R/NR)	Test dataset (R/NR)	Data type	Number of features
GBM	210(140/70)	105(69/36)	105(71/34)	mRNA	17813
				miRNA	533
				SCNV	106
LUSC	121(34/87)	61(17/44)	60(17/43)	mRNA	20530
				miRNA	1045
				SCNV	114

2.3 Dimensionality Reduction via Graph Structure Learning

Let $\mathcal{X} \subset \mathbb{R}^D$ be the input space and $\mathcal{D} = \{x_1, \cdots, x_N\} \subset \mathcal{X}$ be a given dataset. $\mathcal{G} = (\mathcal{V}, \varepsilon)$ represents an undirected graph, where $\mathcal{V} = \{V_1, \cdots, V_N\}$ is a set of vertices and ε is a set of edges. The weight of edges ε is represented by $v_{ij} = \exp(-\|x_i - x_j\|^2/2\sigma^2)$. Suppose that every vertex V_i $(1 \leq i \leq N)$ corresponds to point $z_i \in \mathcal{Z} \subset \mathbb{R}^d$, which lies in a structure with an intrinsic dimension d. We consider learning a function $f_\mathcal{G} \in \mathcal{F}$ such that latent points $\mathcal{Z} = \{z_1, \cdots, z_N\}$ in d-dimensional space could represent $\mathcal{X}$, and $\mathcal{F} = \{f_\mathcal{G} : \mathcal{Z} \to \mathcal{X}\}$ is a set of function defined on a graph $\mathcal{G}$ where function $f_\mathcal{G}$ maps z_i to $f_\mathcal{G}(z_i)$. Our aim is to reduce the dimensionality by learning an intrinsic graph structure in a low-dimensional space. We formulate an optimization problem to minimize the following objective function:

$$\min_{\mathcal{G} \in \widehat{\mathcal{G}}_b} \min_{f_\mathcal{G} \in \mathcal{F}} \{\} \min_{z_1, \cdots, z_N} \sum_{i=1}^{N} \|\mathbf{x}_i - f_\mathcal{G}(\mathbf{z}_i)\|^2 + \frac{\lambda}{2} \sum_{(V_i, V_j) \in \mathcal{E}} b_{i,j} \|f_\mathcal{G}(\mathbf{z}_i) - f_\mathcal{G}(\mathbf{z}_j)\|^2 \quad (1)$$

where the first part is the data reconstruction error and the second part is the reverse graph embedding. In this analysis, $\widehat{\mathcal{G}}_b$ is a feasible set of graphs with a set of vertices $\mathcal{V}$, $\lambda > 0$ is a tradeoff parameter between data reconstruction error and reverse graph embedding, $b_{i,j}$ is the weight of edge (V_i, V_j), and $\mathcal{E}$ is the edge set formed with the edge weight $b_{i,j}$.

2.4 Tree Based Dimensionality Reduction

To avoid the difficulty of learning a general graph, we investigate a family of tree structures for study. The minimum spanning tree is used as the feasible tree graph structure and could be solved by Kruskal' algorithm.[35] A linear projection model $f_\mathcal{G}(\mathbf{z}) = \mathbf{W}\mathbf{z}$ could be used as the mapping function. The following specific formulation is instantiated from (1):

$$\min_{W,Z,B} \sum_{i=1}^{N} \|\mathbf{x}_i - \mathbf{W}\mathbf{z}_i\|^2 + \frac{\lambda}{2} \sum_{i,j} b_{i,j} \|\mathbf{W}\mathbf{z}_i - \mathbf{W}\mathbf{z}_j\|^2 \quad (2)$$

$$\text{s.t. } \mathbf{W}^T\mathbf{W} = \mathbf{I}, \mathbf{B} \in \mathcal{B}$$

where $\mathbf{W} = [\mathbf{w}_1, \cdots, \mathbf{w}_d] \in \mathbb{R}^{D\times d}$ is an orthogonal set of d linear basis vectors $w_l \in \mathbb{R}^D$, $\mathbf{Z} = [\mathbf{z}_1, \cdots, \mathbf{z}_N] \in \mathbb{R}^{d\times N}$ is represented by the projected data points of D in the low dimensional space R^d and $\mathbf{B} = [b_{i,j}] \in \mathbb{R}^{N\times N}$ is an adjacent matrix of a tree $\mathcal{T} = \varepsilon_\mathcal{T}$. In this analysis, $\mathcal{T}$ is a tree with the minimum total cost and $\varepsilon_\mathcal{T} = \{(i,j) : b_{i,j} \neq 0\}$.

2.5 Discriminative Dimensionality Reduction via Learning a Tree

Although data points in a high-dimensional space are projected to latent points in the low-dimensional space in a form of a tree structure, this projection achieves tree-structured data without applying clustering information. To fully utilize clustering data, we focus on a latent points $\{y_k\}_{k=1}^K$ as the center of $\{z_i\}_{i=1}^N$ and formulate the following optimization problem:

$$\min_{W,Z,B,Y,R} \sum_{i=1}^{N} \|\mathbf{x}_i - \mathbf{W}\mathbf{z}_i\|^2 + \frac{\lambda}{2} \sum_{k,k'} b_{k,k'} \|\mathbf{W}\mathbf{y}_k - \mathbf{W}\mathbf{y}_{k'}\|^2 \quad (3)$$

$$+\gamma \sum_{k=1}^{K} \sum_{V_j \in S_k} \|\mathbf{z}_j - \mathbf{y}_k\|^2$$

$$\text{s.t. } \mathbf{W}^T\mathbf{W} = \mathbf{I}, \mathbf{B} \in \mathcal{B}$$

where the third term is the objective function of K-means, Sopf;$= \{S_1, \cdots, S_K\}$ is a partition of $\{V_1, \cdots, V_N\}$, and $\gamma \geq 0$ is a tradeoff parameter between the objective function of the first two terms and empirical quantization error of latent points.

Actually, clustering is imposed on data set using K-means partition algorithm with several drawbacks. Firstly, parameter K is data-dependent and it is very difficult to set it properly. Secondly, clusters in K-means do not always converge into the global optimum. Thus, the hard partition K-means is replaced with a relaxed regularized empirical quantization error given by:

$$\min_{W,Z,B,Y,R} \sum_{i=1}^{N} \|\mathbf{x}_i - \mathbf{W}\mathbf{z}_i\|^2 + \frac{\lambda}{2} \sum_{k,k'} b_{k,k'} \|\mathbf{W}\mathbf{y}_k - \mathbf{W}\mathbf{y}_{k'}\|^2 \quad (4)$$

$$+\gamma[\sum_{k=1}^{K}\sum_{i=1}^{N} r_{i,k}\|\mathbf{z}_i-\mathbf{y}_k\|^2+\sigma\Omega(\mathbf{R})]$$

$$\text{s.t. } \mathbf{W}^T\mathbf{W}=\mathbf{I},\ \mathbf{B}\in\mathcal{B},\ \sum_{k=1}^{K} r_{i,k}=1,\ r_{i,k}\ \geq\ 0,\forall i,\forall k$$

where $\mathbf{R}\in\mathbb{R}^{N\times K}$ with the (i,k)th entry as $r_{i,k}$, $\Omega(\mathbf{R})=\sum_{i=1}^{N}\sum_{k=1}^{K} r_{i,k}\log r_{i,k}$ is the negative entropy regularization, and $\sigma>0$ is the regularization parameter. The negative entropy regularization transforms hard assignment used in K-means to soft assignment used in Gaussian mixture models for further analysis.

2.6 Optimization Algorithm of DDRTree

2.6.1 Fix B, R and Solve W, Z, Y

Given $\mathbf{B}$, $\mathbf{R}$, problem (4) with respect to $\mathbf{W}$, $\mathbf{Z}$, $\mathbf{Y}$ has the following analytic solution:

$$\mathbf{W}=\mathbf{U}(:,1:\mathbf{d}) \tag{5}$$

$$\mathbf{Z}=\mathbf{W}^T\mathbf{XQ} \tag{6}$$

$$\mathbf{Y}=\mathbf{ZR}\left(\frac{\lambda}{\gamma}\mathbf{L}+\Gamma\right)^{-1} \tag{7}$$

where $\mathbf{d}$ is the number of reduced dimension, $\mathbf{Q}=\frac{1}{1+\gamma}[\mathbf{I}+\mathbf{R}\left(\frac{1+\gamma}{\gamma}\left(\frac{\lambda}{\gamma}\mathbf{L}+\Gamma\right)-\mathbf{R}^T\mathbf{R}\right)^{-1}\mathbf{R}^T]$, $\Gamma=\mathrm{diag}\left(1^T\mathbf{R}\right)$ and the Laplacian matrix over a tree encoded in $\mathbf{B}$ is defined as $\mathbf{L}=\mathrm{diag}(\mathbf{B}1)-\mathbf{B}$ respectively. In this analysis, $\mathbf{W}$ is an orthogonal set of d linear basis vectors, $\mathbf{Z}$ is the reduced dimension space tree which represents the smooth tree graph embedded in the low dimensional space, and $\mathbf{Y}$ represents the latent points as the center of $\mathbf{Z}$. We also set $\mathbf{C}=\mathbf{XQX}^T$ and performed eigen-decomposition on $\mathbf{C}$ such that $\mathbf{C}=\mathbf{U}\Lambda\mathbf{U}^T$, where $\mathbf{U}$ and $\mathrm{diag}(\Lambda)$ are the eigenvectors and eigenvalues of matrix with sorted in a descending order respectively.

2.6.2 Fix W, Z, Y and Solve B, R

Given $\mathbf{W}$, $\mathbf{Z}$, $\mathbf{Y}$, problem (4) with respect to $\mathbf{B}$, $\mathbf{R}$ is summarized as follows:

$$\min_{\mathbf{B}\in\mathcal{B},\mathbf{R}} \frac{\lambda}{2}\sum_{k,k'} b_{k,k'}\|\mathbf{W}y_k-\mathbf{W}y_{k'}\|^2+ \gamma[\sum_{i=1}^{N}\sum_{k=1}^{K} r_{i,k}\|\mathbf{z}_i-\mathbf{y}_k\|^2+\sigma\sum_{i=1}^{N}\sum_{k=1}^{K} r_{i,k}\log r_{i,k}] \tag{8}$$

$$\text{s.t. } \sum_{k=1}^{K} r_{i,k}=1,\ r_{i,k}\ \geq\ 0,\forall i,\forall k$$

We have the optimal analytic solution $\mathbf{R}$ given by the following form:

$$r_{i,k}=\exp(-\|\mathbf{z}_i-\mathbf{y}_k\|^2/\sigma)/\sum_{k=1}^{K}\exp(-\|\mathbf{z}_i-\mathbf{y}_k\|^2/\sigma),\quad \forall i,\forall k \tag{9}$$

To obtain the optimal $\mathbf{B}$, the optimization problem (8) is represented by $\min_{\mathbf{B}\in\mathcal{B}}\sum_{k,k'} b_{k,k'}\|y_k-y_{k'}\|^2$, which could be solved by Kruskal's algorithm.

2.7 Dimensionality Reduction of DDRTree

In this analysis, an R implementation of the DDRTree available in the DDRTree package was used for dimensionality reduction.[35] DDRTree has two parameters d and λ, where d is the number of reduced dimension and λ is the regularization parameter for reverse graph embedding. These two parameters could be set empirically where parameter d and parameter λ were varied among the candidate set $\{10,20,30,40,50\}$ and $\{2^{-2},2^{-1},2^{0},2^{1},2^{2}\}$ respectively to form different parameter combinations. The smooth tree graph embedded in the low dimensional space was derived based on various parameter combinations and further employed as input features for classifier development.

2.8 Evaluation of SVM Classifier

An R implementation of the SVM available in the e1071 package was employed for SVM model development, and the Gaussian kernel function was used. SVM has two parameters c and γ_{SVM}, where c is the soft margin parameter and γ_{SVM} is the parameter for the radial basis kernel function. For tuning the parameters, we let each of them vary among the candidate set $\{10^{-2},10^{-1},10^{0},10^{1},10^{2}\}$ to form different parameter combinations. Each combination of parameter choices was evaluated using 5-fold cross-validation, and the parameters with best cross-validation AUC (the Area under the Receiver Operating Characteristic (ROC) Curve) were identified. The SVM model was then trained on the whole training set based on the optimal parameters, tested on the independent validation dataset, and evaluated based on both AUC and accuracy.

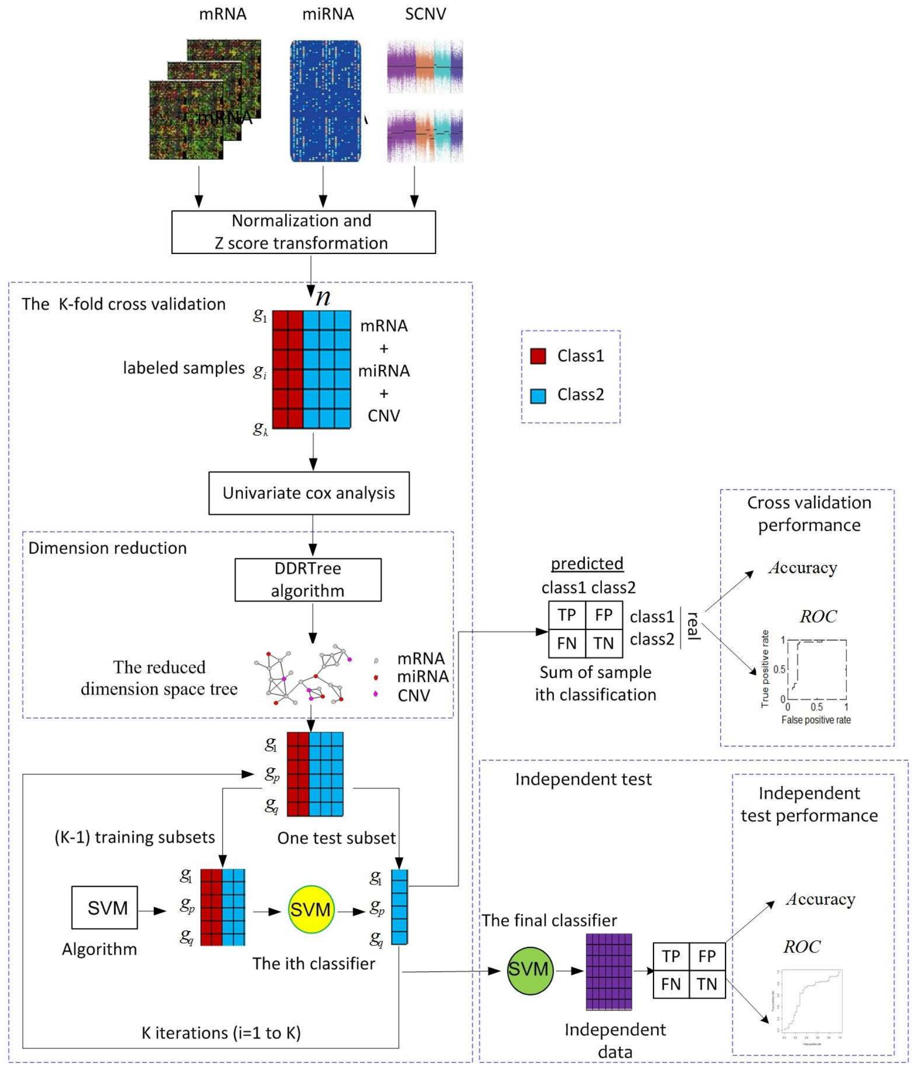

Figure 1. Workflow for the development and evaluation of the tree based SVM classifier for cancer recurrence prediction. Multiple omics data of cancer genome-wide gene expression, microRNA expression and somatic copy number alteration profiles are collected, normalized and then z-score transformed. The univariate Cox analysis is used to preselect the multi-markers for dimensionality reduction of multi-omics data. DDRTree algorithm is then applied to dimensionality reduction for learning a tree structure. A K-fold cross-validation is used for SVM classifier development. Fully developed classifier based on the optimal parameters identified in cross-validation is then evaluated in an independent test data set.

3 Results

3.1 Overview of Cancer Recurrence Prediction Workflow

Figure 1 illustrates the overview of tree-based dimensionality reduction for the integrative analysis of multiple molecular data and SVM classifier development. Multiple omics datasets such as gene expression, microRNA expression and somatic copy number alteration profiles on a specific cancer type are collected and preprocessed for integrative analysis. The univariate Cox proportional hazard regression is then performed to estimate the p value for each feature of molecular data. The identified multi-markers are associated with overall survival of cancer patients with statistical significance ($p < 0.05$). DDRTree achieved different smooth tree graphs based on various parameter combinations which were then used for developing the outcome prediction model. Accordingly, SVM classifier was proposed for cancer recurrence prediction with a K-fold cross-validation. Data on selected features from smooth tree graph is used to parameterize the SVM classification algorithm. The parameterized classifier (yellow circle) is then applied to the test subset for classifying samples. The cross-validation process is repeated with K times and the K results from the subsets then could be united to realize a single estimation. Fully classifier (green circle) was developed with the optimal parameters discovered by the AUC value and then validated on an independent data set.

3.2 DDRTree Achieved Reduced Dimension Space Tree with Statistical Significance

The univariate Cox proportional hazard regression analysis was applied to multiple omics data for estimating the p value of each feature from molecular data on survival time and then discover multi-markers. In this analysis, an panel of 3398 multi-markers ($p < 0.05$; mRNA,3261; miRNA,127; SCNA,10) were significantly associated with overall survival in the training dataset of GBM. On the other hand, a total of 4193 multi-markers with p value less than 0.05 were considered as significant features in the training dataset of LUSC, including 3768 mRNAs, 407 miRNAs and 18 SCNAs respectively.

We performed dimensionality reduction method DDRTree on the molecular data with 3398 multi-markers in the training dataset of GBM. To avoid the difficulty of tuning too many parameters, we fixed $\gamma = 10$ and $\sigma = 0.01$ respectively for study in all the experiments. The parameter values including the number of reduced dimension ($d = 30$) and the regularization parameter for reverse graph embedding ($\lambda = 1$) were then set with an optimal reduced dimension space tree selection. A scatter plot was generated for average d-dimensional expression value of each sample in the recurrence and non-recurrence patients. P value was calculated using Welch's t-test between recurrence and non-recurrence cohorts, and that less than 0.05 was considered statistically significant. As shown in Supplementary Figure S1a–b, a scatter plot showed statistical significance in the training dataset ($p = 9.66e-07$) and test data set ($p = 2.17e-06$) respectively. On the other hand, DDRTree algorithm was applied to LUSC with the optimized parameters $d = 10$ and $\lambda = 1$ respectively. As can be seen from Supplementary Figure S1c–d, it illustrated that a scatter plot displayed a significant test for the training dataset ($p = 0.01$) and test data set ($p = 0.02$) respectively.

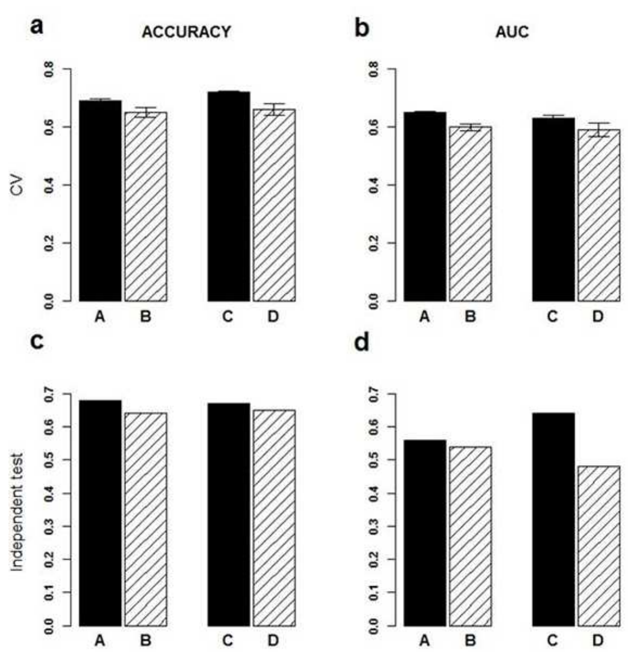

Figure 2. Performance of tree based and t-test based SVM classifiers predicting cancer recurrence. (a) Accuracy from 5-fold cross-validation results with the training dataset of each cancer type. (b) AUC from 5-fold cross-validation results with the training dataset of each cancer type. (c) Accuracy of independent validated test. (d) AUC of independent validated test. Data are presented as the mean ± variance in (a) and (b). A: DDRTree for GBM, B: t-test for GBM, C: DDRTree for LUSC, D: t-test for LUSC.

3.3 Tree Based SVM Classifier Improves Prediction Performance of Cancer Recurrence

It is generally believed that the integration of multi-omics data improves the power in predicting clinical outcomes. Therefore, we first investigated whether a tree based framework for integrating molecular data could improve prediction performance of clinical outcomes in GBM as an example. For each combination, results were generated for five random samples when possible in order to obtain robust performance evaluation results. For reference, we compared performance from DDRTree to that from the commonly used method t-test. For each random sample, t-test was used for feature selection with a cutoff value of $p = 0.05$.

Figure 2a–b depicts the average accuracy and AUC of tree and t-test based SVM classifiers from the cross-validation studies in GBM. Tree based SVM achieved 0.69 in accuracy and 0.65 in AUC respectively. Tree based SVM achieved 6.2% increase in accuracy and 8.3% increase in AUC as compared to t-test based SVM classifier respectively. Figure 2a–b also shows small variance across the five random samples for each combination, suggesting the robustness of the evaluation results. Using parameters selected based on cross-validation results, full models were developed based on the training dataset and tested on the independent test data set. The tree based SVM classifier achieved an accuracy of 0.68, which was 6.3% higher than that from the t-test based SVM classifier (Figure 2c). Similarly, the tree based SVM classifier obtained an AUC of 0.56, which was 3.7% higher than that from the t-test based SVM classifier (Figure 2d).

To further evaluate the effectiveness of the tree-based dimensionality reduction approach for clinical outcome prediction, we reversed the training and testing datasets by constructing SVM classifier based on the multiple molecular data in GBM. Tree based SVM yielded 0.71 in accuracy and 0.68 in AUC respectively from the cross-validation studies. The SVM classifier was developed using the optimal parameters and then validated on the test data set. The tree based SVM classifier achieved an accuracy of 0.65 and an AUC of 0.62 respectively. The performance proved that

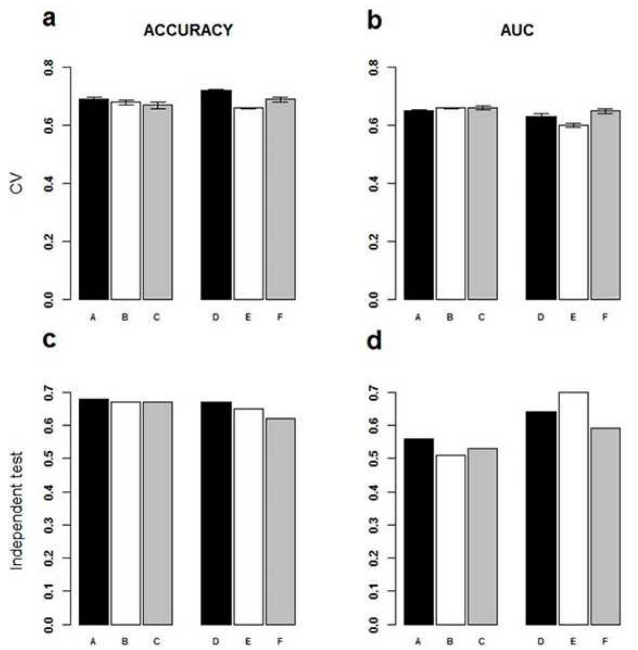

Figure 3. Performance of tree based SVM classifier with different p value cutoffs including 0.05, 0.01, and 0.005 for discovering the multi-maker candidates based on the Cox proportional hazard model. (a) the accuracy of 5-fold cross-validation results with the training dataset of each cancer type. (b) the AUC from 5-fold cross-validation results with the training dataset of each cancer type. (c) the accuracy of independent test data set. (d) the AUC of independent test data set. Data are presented as the mean± variance in (a) and (b). A: GBM, B: GBM, C: GBM, D: LUSC, E: LUSC, F: LUSC. Black: cutoff 0.05, White: cutoff 0.01, Gray: cutoff 0.005.

the tree based SVM classifier is powerful for clinical outcome prediction.

To test the general applicability of the tree based SVM classifier, we applied it to LUSC dataset for predicting clinical outcomes. The training dataset of LUSC with 61 samples was used to develop classifiers for the prediction of 3-year recurrence. In cross-validation studies, tree based SVM classifier achieved 9.1% increase in accuracy (0.72) as compared to *t*-test based SVM classifier (Figure 2a). When AUC (0.63) was used as the performance metric, 6.8% increase were observed as compared to *t*-test based SVM classifier (Figure 2b). Applying the full model developed by tree based SVM classifier on the independent validation dataset resulted in an accuracy of 0.67, which was 3.1% higher than that obtained by *t*-test based SVM classifier (Figure 2c). On the other hand, tree based SVM classifier achieved an AUC of 0.64, which was 33.3% increase as compared to *t*-test based SVM classifier (Figure 2d). Both cross-validation results and independent test results clearly demonstrated that tree based SVM classifier improved the predicting power and achieved significantly better performance for clinical outcome prediction.

3.4 Tree Based SVM Classifier is Robust to the Different Number of Multi-markers

We used DDRTree with the different multi-markers from the univariate Cox proportional hazard regression analysis and achieved the various reduced dimension space trees. In order to evaluate the robustness of tree for the selected features in difference, we tested different *p* value cutoffs including 0.05, 0.01, and 0.005.

Using parameters selected based on cross-validation results, full models were developed based on the training dataset in GBM and then tested on the independent test data set. According to both accuracy and AUC, performance based on the *p* value cutoffs 0.05, 0.01, and 0.005 were very similar (Figure 3a–c). Interestingly, when AUC was used as the performance metric, the performance based on the cutoff 0.05 was 9.8% higher than that obtained based on cutoff 0.01 and 5.7% higher than that obtained based on cutoff 0.005 respectively (Figure 3d).

To test the general robustness of the tree based SVM classifier for different types of cancer, we applied it to LUSC dataset for predicting clinical outcomes. The accuracy was very similar in cross-validation (Figure 3a) and test data set (Figure 3c) based on the different *p* value cutoffs. The performance metric AUC was very similar in cross-validation based on three *p* value cutoffs (Figure 3b). In the test data set,AUC value based on the cutoff 0.01 was 9.4% greater than that based on the cutoff 0.05 and 18.6% greater than that based on the cutoff 0.005 respectively (Figure 3d). These results indicate that tree based SVM classifier was reasonably robust to the different picked features with different *p* value cutoffs. Because the 0.05 cutoff seemed to perform slightly better than its more stringent alternatives, we used the 0.05 cutoff for further study.

3.5 Combining Multiple Omics Data Improves Prediction Performance of Cancer Recurrence

The Cancer Genome Atlas (TCGA) studies have characterized key genetic variations and discovered therapeutic strategies through generating genomic, transcriptomic, epigenomic and proteomic data from a large number of cancer samples. It is of particular interest to investigate whether integration of multi-omics data could lead to improved prognosis, and how performance changes with the integration of molecular data.

According to the multi-markers from multiple types of molecular data, we tried to dissect their individual contribution to the observed performance. Similar to the above analysis, we performed DDRTree on the molecular data with 3261 mRNAs, 127 miRNAs, and 10 SCNAs respectively in GBM. The smooth tree graphs were obtained with various parameter combinations and further applied to input features for classifier development. As shown in Figure 4a–b, smooth tree graphs derived from mRNA, miRNA, and

SCNA respectively did not result in comparable performance as that from the integration of molecular data. The accuracy for the integrated molecular data based classifier on independent validation set was 0.68, which was 13.3% higher than that for the mRNA-based classifier, 25.9% higher than that for the miRNA-based classifier and 19.3% higher than that for the SCNA-based classifier respectively (Figure 4a). On the other hand, the AUC was achieved by integrated molecular data based classifier was 9.8% greater than that by mRNA-based classifier, 12% greater than that by miRNA-based classifier and 14.3% greater than that by the SCNA-based classifier respectively (Figure 4b).

To further evaluate the applicability of the integration of molecular data, the SVM classifier was also developed in LUSC. To acquire the smooth tree graphs, DDRTree was applied to the molecular data with 3768 mRNAs, 407 miRNAs, and 18 SCNAs respectively in the training dataset of LUSC. As shown in Figure 4c–d, improvement in accuracy and AUC on independent validation set was observed with the integrated molecular data based SVM classifier. The accuracy and AUC were 0.67 and 0.64 respectively with the integrated molecular data based SVM classifier, which were 11.7% and 8.5% higher than those with mRNA-based SVM classifier, 17.5% and 16.4% higher than those with miRNA-based SVM classifier, and 11.7% and 30.6% higher than those with SCNA-based SVM classifier respectively (Figure 4c–d). These results demonstrated that multiple types of omics data could help improve prediction performance.

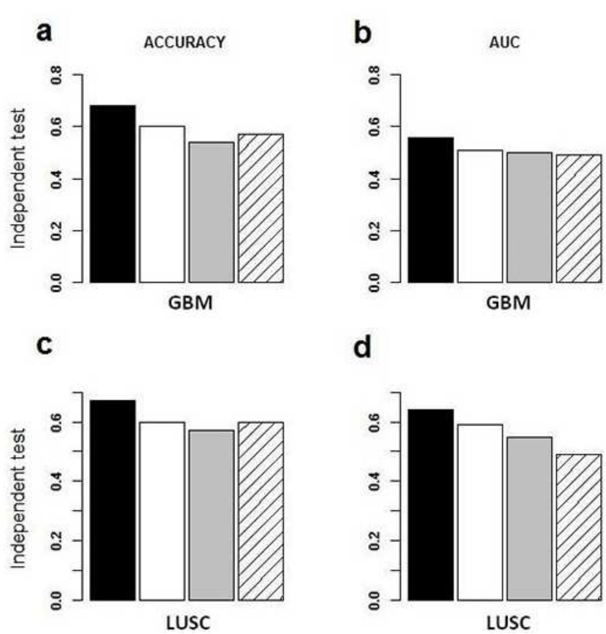

Figure 4. The predicted outcome of independent test data set by tree based SVM classifier with the integrated molecular data and each type of molecular data independently. (a) the accuracy of test data set in GBM. (b) the AUC of test data set in GBM. (c) the accuracy of test data set in LUSC. (d) the AUC of test data set in LUSC. Black: multi-omics data of mRNA, miRNA and SCNV, White: mRNA, Gray: miRNA, Twilled white: SCNV.

3.6 Tree Based SVM Classifier Achieves Superior Prediction Performance

To further test the effectiveness of the proposed method, we compared DDRTree with feature selection methods including principal component analysis (PCA),[15] the univariate Cox proportional hazard regression analysis,[36] Lasso[37] and Recursive Feature Elimination (RFE).[38] We conducted PCA from multiple types of molecular data and principal components were chosen as selected features based on their cumulative percentage of variance higher than 90%. Similar to the above analysis, the univariate Cox proportional hazard regression was utilized to identify the multi-markers which were significantly associated with overall survival of cancer patients ($p < 0.05$). In this analysis, an R implementation of the Lasso available in the glmnet package was employed for feature selection. On the other hand, Recursive Feature Elimination (RFE) for support vector machines was utilized for multivariate feature selection.[38] SVM-RFE performed feature selection by iteratively training a SVM classifier with the current set of features and removing the least important feature.

As shown in Figure 5a–b, tree based SVM classifier achieved 1.5% increase in accuracy and 3.7% increase in AUC as compared to PCA based SVM classifier for independent test data set in GBM. On the other hand, tree based SVM classifier attained 4.6% increase in accuracy and 16.7% increase in AUC respectively in comparison with the univariate Cox proportional hazard regression analysis. Compared to the performance of Lasso based SVM, tree based SVM classifier achieved 7.9% higher accuracy and 14.2% higher AUC. The accuracy and AUC were 0.66 and 0.55 respectively with RFE based SVM classifier, which were 2.9% and 1.8% lower than those with Tree based SVM classifier (Figure 5a–b).

Applying tree based SVM classifier on the independent validation set resulted in an accuracy of 0.67 and AUC of 0.64 in LUSC, which were 1.5% and 4.9% higher than those obtained by PCA, 8.1% and 8.5% higher than those obtained by the univariate Cox proportional hazard regression analysis, 11.6% and 12.2% higher than those obtained by Lasso, and 3.1% and 1.6% higher than those obtained by RFE respectively (Figure 5c–d). These results demonstrated that tree based SVM classifier achieved similar or better prediction performance as compared to state-of-the-art feature selection methods.

4 Discussion

In contrast to earlier researches driven by a single molecular data type (mRNA, microRNA, or somatic copy number

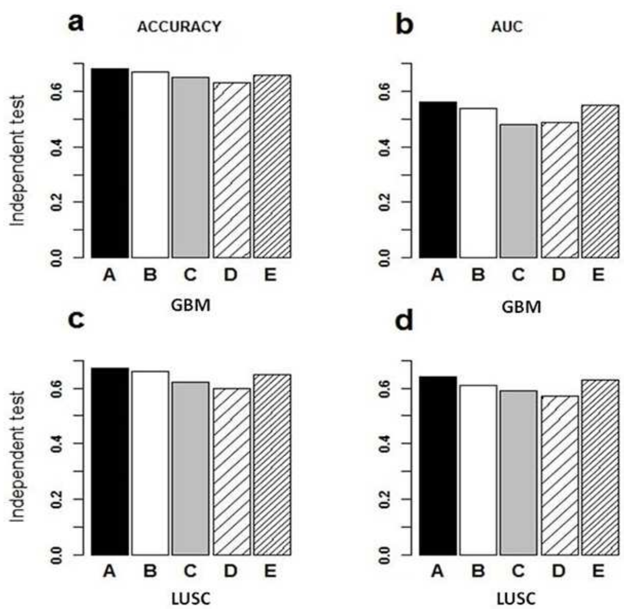

Figure 5. The outcome of five different types of methods for independent validation dataset. (a) the accuracy of independent test data set in GBM. (b) the AUC of independent test data set in GBM. (c) the accuracy of independent test data set in LUSC. (d) the AUC of independent test data set in LUSC. A: DDRTree, B: PCA, C: univariate Cox proportional hazard regression analysis, D: Lasso, E: RFE.

alteration), we systematically evaluated clinical outcome prediction from the integration of the knowledge on mRNA, microRNA, and somatic copy number alteration and described its potential prognostic relevance across multiple cancer types. The analysis goal is to apply tree-based dimensionality reduction method for clinical outcome prediction, with particular focus on the predictive power of reduced dimension space tree. As compared to state-of-the-art feature selection methods, the tree based SVM classifier successfully achieved similar or better performance of cancer recurrence prediction. Our results demonstrated great potential of DDRTree in clinical outcome prediction.

In our analysis, the tree-based dimensionality reduction approach used for the integrative analysis of multiple molecular data has four distinct advantages. Firstly, it offered a promising method of feature space dimensionality reduction. For the small sample size problem in gene expression study, reducing the dimensionality of gene expression data is critical for clinical outcome prediction. Although statistically important genes of reduced dimension space tree may not be related with expression variance in a tumor-specific sample, multiple genomics expression alterations among these genes are probably to cause cancer-associated phenotype changes. Secondly, tree structure based optimized parameters with cross-validation was investigated to reason the rich genotype information embedded in tree-structured data. Specifically, tree-structured data usually contain both topological information and certain attributes associated with each tree node or edge.[30] Thirdly, the tree-based dimensionality reduction method was developed to obtain the compact feature representation for clustering problems.[35] By classifying patterns into different groups in an unsupervised manner, tree-structured data clustering could predict clinical outcome of cancer. Fourthly, it is generally believed that the spaces where the tree-structured data lie in could be strongly non-Euclidean,[39] and tree-structured data thus recover an explicit smooth graph structure with local geometry only captured by distances of data points in the low dimension space.[35] Moreover, the tree-based structures provided the insight about the relationships with the cancer recurrence.[40,41] Specifically, tree-based structures implied the rich genotype information embedded in tree-structured data which were correlated with cancer-associated phenotype changes.

Early single-omic data aimed to obtain a single "layer" of information from cancer patients and have been widely used for clinical outcome prediction.[7,42] Recently, a number of multi-omics approaches have been applied to predict survival and relapse of cancer.[11,12,43] The availability of such resources is at the core of generating data-driven hypotheses for cancer research. Integration of different omics data types is typically used to discover potential mutations that lead to cancer relapse, and treatment targets that could be then tested in clinical cancer research. Moreover, these molecular data could be useful to give insight to which biological pathways are critical for cancer pathogenesis. In our analysis, the results suggest that the tree-structured data from the integrative analysis of multiple molecular data is significant for achieving the observed performance. The performance suggested that multiple types of omics data could help improve prediction performance, but the increase in accuracy using mRNA expression data was relatively smaller than miRNA expression data and somatic copy number alteration data (Figure 4). The possible explanation is the filtered data was not from a threshold of the differential expression values in miRNA and SCNA data but from the significance P-value in the Cox regression approach. Namely, too many gene expression values of small "effect size" might be drowning out the contribution of the relevant data in the general noise level. Furthermore, DDRTree achieves reduced dimension space tree, which is different from the Double Kernel-Based Clustering method for Gene Selection (DKBCGS) in expression data analysis.[44]

The R implementation of the random forest (RF) algorithm was used for study as compared to SVM. RF algorithm has two parameters ntree and mtry, which could be set empirically for training the RF classifier. We tuned the ntree among the candidate set $\{10^1, 10^2, 10^3\}$ and mtry among the candidate set $\{2^1, 2^2, 2^3, 2^4, 2^5\}$ respectively to form fifteen different parameter combinations. Each combination of parameter choices was evaluated using 5-fold cross-validation, and the parameters with best cross-validation AUC were identified. The final classifier was then trained on the whole training set using the optimal parameters. The developed RF classifier was validated on the independent validation dataset, and the classification performance was evaluated with AUC and accuracy. In the cross-validation analysis for GBM patients, tree based RF classifier yielded an AUC of 0.63, which was 3% lower than that obtained by tree based SVM classifier. Moreover, it achieved an accuracy of 0.63, which was 8.7% lower than that obtained by tree based SVM classifier. When applying the RF classifier on the independent validation dataset, it achieved an AUC of 0.56, which was similar with that obtained by tree based SVM classifier. On the other hand, tree based RF classifier achieved an accuracy of 0.64, which was 5.9% lower than that obtained by tree based SVM classifier. The results indicated that SVM classifier achieved clearly superior performance compared to RF classifier.

A variable importance evaluation method has been proposed where the number of selected features is determined by optimality criteria rather than arbitrarily by the user like a t-test.[45] An R implementation of the *var.select* in the randomForestSRC package was utilized for finding variable importance values of mRNA, miRNA and SCNA in GBM patients. Variable importance (VIMP) evaluates the predictive performance of the above features and a large VIMP value indicates a potentially predictive variable. In this analysis, the corresponding VIMP values for an panel of 547 multi-markers were not equal to 0 in the training dataset of GBM. We performed dimensionality reduction method DDRTree on the molecular data with 547 multi-markers in the training dataset of GBM. Tree based SVM yielded 0.62 in accuracy and 0.65 in AUC respectively from the cross-validation studies. The SVM classifier was developed using the optimal parameters and then validated on the test data set. The tree based SVM classifier achieved an accuracy of 0.59 and an AUC of 0.54 respectively. These results demonstrated that tree based SVM classifier based on 3398 multi-markers achieved similar or better prediction performance as compared to that based on 547 multi-markers.

Our study has several limitations. First, it is a common problem for overfitting in supervised machine learning due to the multiple-omics data with small sample size. In this analysis, the performance was notably worse than that estimated based on cross-validation when the tree based SVM classifier was applied on an independent data (Figure 2). On the one hand, the explanation is that the different class proportions are observed between the training and the independent test data set. On the other hand, the Z-score transformation across all samples maybe result in overfitting due to the fact that all training and test samples have the same distribution. Moreover, the pre-filtering step using an external method (univariate Cox proportional hazard regression analysis) could also lead to the over-optimistic results. Second, performance such as accuracy and AUC have been widely used for evaluating the predictive power in machine learning. Actually, it has been noticed that accuracy and AUC may not always coincide and even be in contradiction with one another.[46] In our study, we also noticed clear difference between performance gain in AUC and accuracy. Third, similar to conventional dimensionality reduction methods like PCA and ICA, the reduced dimension space trees are derived from existing multiple molecular data, and then need to be recomputed for the added new data. Finally, when applying the SVM classifier on the independent validation set of GBM, the tree based SVM classifier achieved an accuracy of 0.7(50/71) for recurrent samples and 0.62(21/34) for non-recurrent samples respectively. When applying the SVM classifier on the independent validation set of LUSC, the tree based SVM classifier achieved an accuracy of 0.47(8/17) for recurrent samples and 0.75(32/43) for non-recurrent samples respectively. The performance indicates that unbalanced dataset tends to be a problem for classifier because the majority class usually produces the good predicted results.

The TCGA has illustrated different types of cancer for the analyses of mRNA, miRNA, copy number alteration, somatic mutation, methylation, and protein expression data. The integration of mRNA, miRNA and SCNV provides the key path to better understand cancer phenotypes, and thereby offers a global perspective of the interplay between the various levels of molecular data. Our results demonstrated that tree based SVM classifier, when incorporated in reduced dimension space tree recognition algorithms, showed great potential in multi-omics data based outcome prediction for cancer patients. Crucially, this study highlights the value that different types of molecular data could be integrated to identify the smooth tree graph as targets and have the great potential to inform patient prognosis.

5 Author Contributions

Conceived and designed the experiments: MS. Performed the experiments: MS JW. Analyzed the data: MS CZ. Contributed reagents/materials/analysis tools: MS. Wrote the paper: MS.

Conflict of Interest

Non declared.

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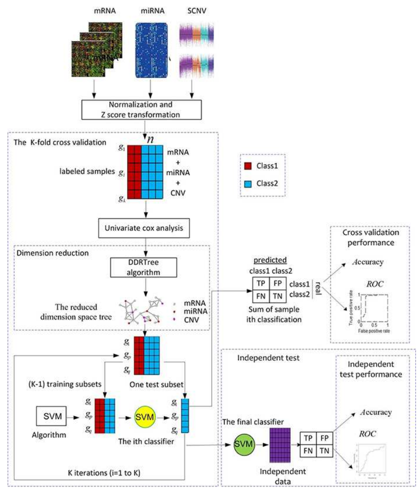

M. Shi, J. Wang, C. Zhang*

1 – 12

Integration of Cancer Genomics Data for Tree-Based Dimensionality Reduction and Cancer Outcome Prediction